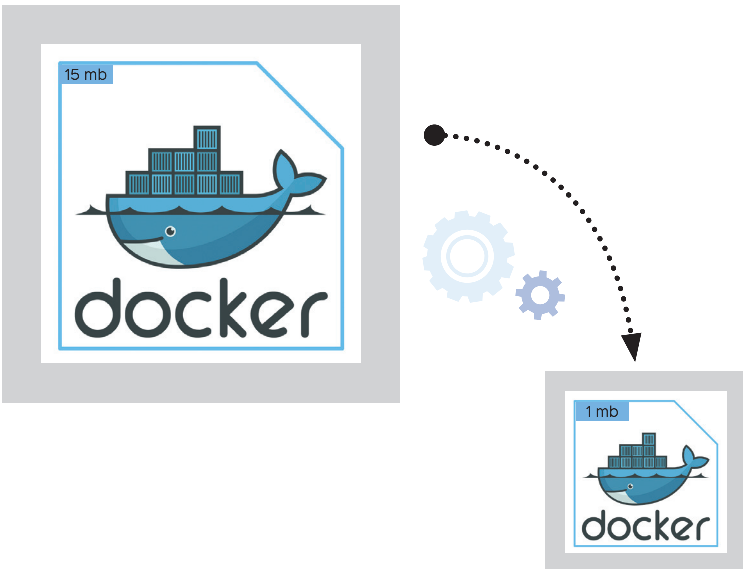
Reducing the Size of Your Docker Image

Why stick with a 6GB Docker image that functions? Downtime's crucial; services require updates and lightweight images mean faster deployments, saving time.



I work on a project called Kube-ez (*www.github.com/kitarp29/kube-ez*). It is a simple Golang project which has an image of nearly 0.5GB in size. This resulted in excessive time consumption while spinning up any container with this image. This issue not only affected deployments but also slowed down the CI jobs, leading to a delay in my entire software development life cycle.

I was able to fix this issue by reducing the image size by 97%, turning it from ~0.5GB to a mere 16MB. This change enabled me to build more reliable deployments and faster CI jobs.

Let me present the approach I adopted to address this problem, and discuss some alternative solutions.

Understanding the problem

In the world of microservices, most production relies on containers. Images act like the CDs/flash drives used to play a certain song (your software) in your container.

Docker images aim to bring unity among the developers despite the platform and environment. However, achieving this should not come at the cost of memory and speed.

- Bulky images take longer to download.
- You need increased time to spin up a container using them, slowing deployments and CI pipelines.
- They increase the load and cost of the image registry.
- With upgrades, it becomes harder to check for vulnerabilities.

Why solve this problem?

I often hear from developers: "So what if the Docker image is 6GB; it still runs, right? Don't touch it."

However, I disagree based on my experience as a site reliability engineer at Juspay (an Indian fintech startup).

Keeping a service up and running is always a significant challenge. But let's agree that no company has an SLO of 100%. Each second of downtime counts!

Someday, the service is bound to crash or run into errors. Even in the best-case scenario, you will need to update it sometime. On that day, when you need to spin up a new container or an updated container, having a lighter Docker image will help.